

GitHub link:

<https://github.com/MTHIRUNEELAKANDAN>

Fraud

Guarding Transactions with AI-powered Credit Card Fraud

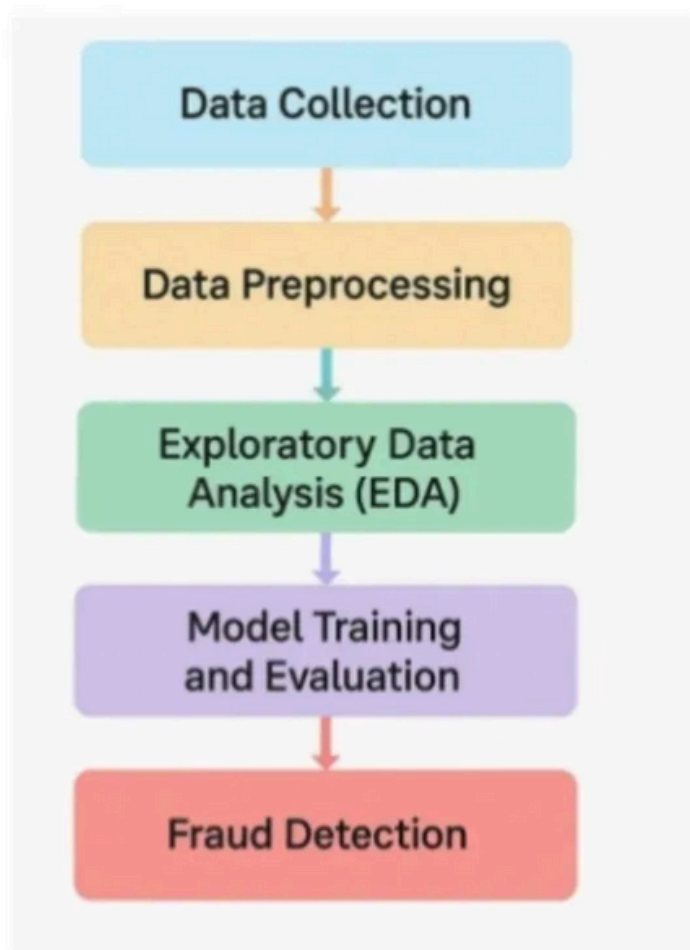
Detection and Prevention

In today's digital economy, credit card transactions are ubiquitous, but so is fraud. Financial institutions face growing challenges in identifying and preventing fraudulent activities in real-time. AI-based fraud detection offers the potential to analyze large volumes of transaction data and detect anomalies effectively.

2. Project Objectives

- Detect fraudulent credit card transactions using machine learning.
- Reduce false positives and improve the accuracy of detection.
- Analyze transaction features to understand fraud patterns.
- Build a model that can predict and prevent future fraudulent activities.

3. Flowchart of the Project Workflow



4. Data Description

The dataset contains anonymized credit card transactions with features such as time, amount, and other variables. The target variable has two class labels indicating fraud. Features are the result of a PCA transformation for confidentiality. Class 1 indicates fraud, and Class 0 indicates legitimate transactions.

5. Data Preprocessing

- Checked for missing values and normalized the 'Amount' feature.
- Performed feature scaling to improve model performance.
- Handled imbalanced data using under-sampling and SMOTE.

6. Exploratory Data Analysis (EDA)

- Analyzed class distribution and correlation between features.
- Visualized fraud vs non-fraud transaction patterns.
- Studied transaction amounts and time gaps for both classes.

7. Feature Engineering

- Engineered time-based and amount-related features.
- Selected important features based on correlation and model importance.
- Applied PCA and autoencoders for dimensionality reduction.

8. Model Building

- Trained models like Logistic Regression, Random Forest, and XGBoost.
- Used cross-validation to improve generalization.
- Evaluated using precision, recall, F1-score, and ROC-AUC.

9. Results & Insights

- Random Forest performed best with high recall for fraud detection.
- SMOTE significantly improved model's ability to detect rare fraud cases.
- A real-time fraud detection system could be built using this model.

10. Tools and Technologies Used

- Python libraries: pandas, numpy, sklearn, matplotlib, seaborn, imbalanced-learn.
- Jupyter Notebook for experimentation.
- Environmental: Google colab

7. Team Members and Roles

- Responsible for data cleaning and EDA:Arulirasan.G
- Leads model building and evaluation:Anandharaman.M
- Handles API development and deployment:Srimanoj.C
- Ensures timelines are met and oversees integration with client systems: Thiruneelakandan